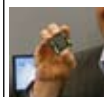


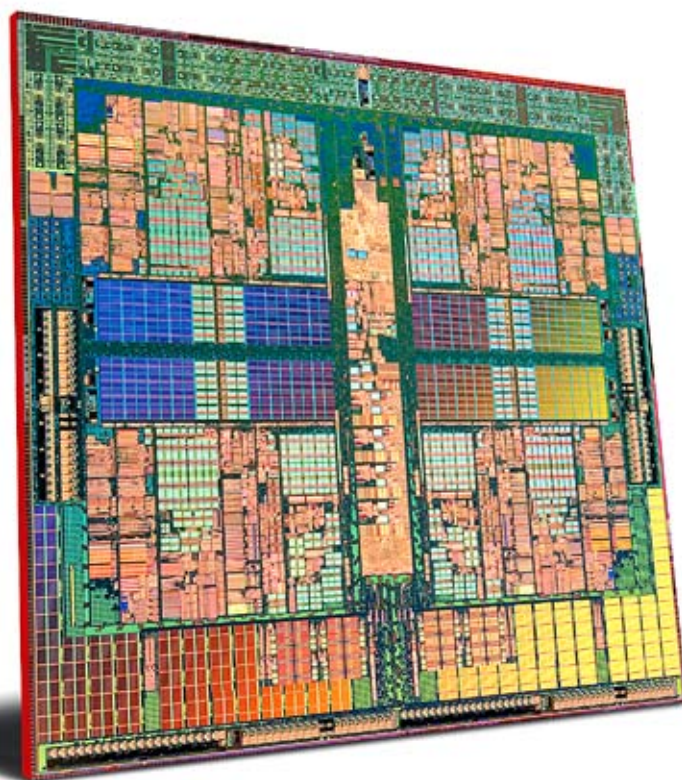


AMD's Athlon 64 was the first processor with an integrated memory controller



Intel will move to a 32nm fabrication with Westmere

AMD touts its Phenom 2 processors as a return to competition. We check the validity of this bold statement



Finally Phenom(enal)?

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Just over a year back, AMD released a new processor family – codenamed *Barcelona* for servers, and *Agena* for desktops. The architecture was codenamed *K10*. A lot of hope was pinned on the resulting Phenom processor range, but as we know they could not keep up with Intel's shiny new dual- and quad-core processors. Intel had a huge advantage with its Core architecture, timely revisions to the fabrication process and tweaks to the CPUs themselves – all of which kept the Core 2 based CPUs well ahead, in nearly all benchmarks.

Intel also moved to a smaller, more efficient 45-nanometer fabrication process and promptly released new dual- and quad-core processors based on it. These were codenamed *Penryn* and quite frankly the performance gap just widened. AMD didn't have an answer to these CPUs and its dwindling presence in the desktop segment

is testament to this fact. The battle between Intel and AMD could not have been more one-sided after Intel debuted its Core i7 architecture late last year. Industry experts believed this to be just one more nail in AMD's coffin and it's no secret that the Nehalem CPUs based on this architecture are currently the fastest desktop processors. Away from the limelight, and according to some, too weak to fight back, AMD quietly released its own 45nm counterparts late last year. These processors were named *Phenom 2* CPUs and the architecture was codenamed *Shanghai*.

AMD has been selling the concept of a computing platform for a while now and this makes sense both in terms of consumers as well as business profitability. Imagine an ecosystem provided by a single manufacturer – consisting of CPU, motherboard chipset and graphics solution. Any such platform can (obviously) be highly optimised and tweaked for maximum possible stability, performance and compatibility since one entity has control over development of all three components.

The first generation platform called *Spider* consisted of an AMD Phenom processor, a 790FX chipset motherboard and an ATi Radeon 3xxx series GPU. The new platform, called *Dragon*, consists of an AMD Phenom 2 processor, a 790FX / 790GX / 790X motherboard and an ATi Radeon 4xxx series GPU.

Architecturally speaking, Phenom 2 is not very different from the original Phenom. The most obvious difference from spec sheets is the huge increase in L3 cache. While Intel's Core i7 has eight MB of L3 cache, the Phenom had just two MB. On the Phenom 2 this has grown to 6 MB, meaning each core gets 1.5 MB of cache although this cache is common to all cores. The L1 and L2 caches remain identical to earlier Phenom processors. The other major advantage that Phenom 2 has is the fact that at 45nm, its die size is a lot smaller than the 65nm die of the Phenom. This allows these processors to scale higher clock speeds and cross the three GHz barrier – a feat Intel achieved awhile back.